

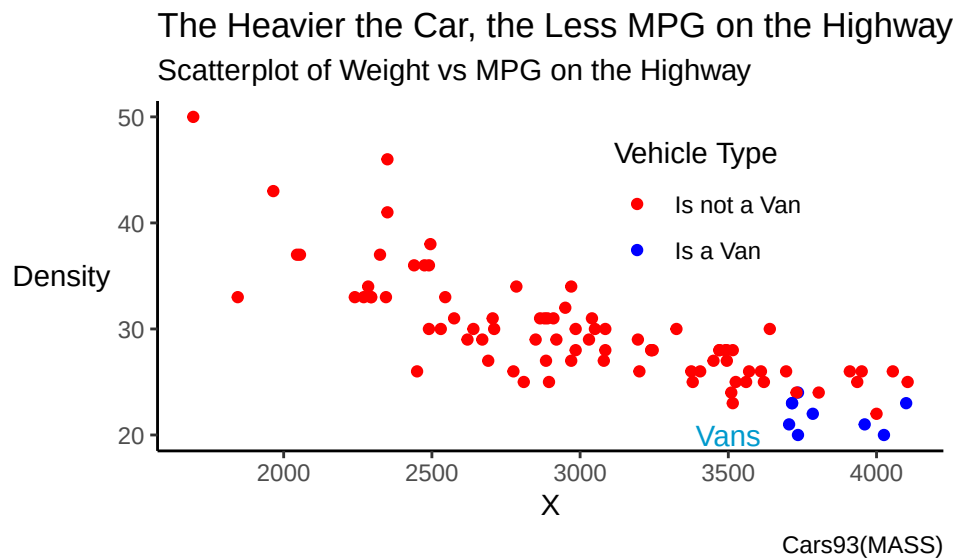
Color Open Ended Assignment

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9/24/25

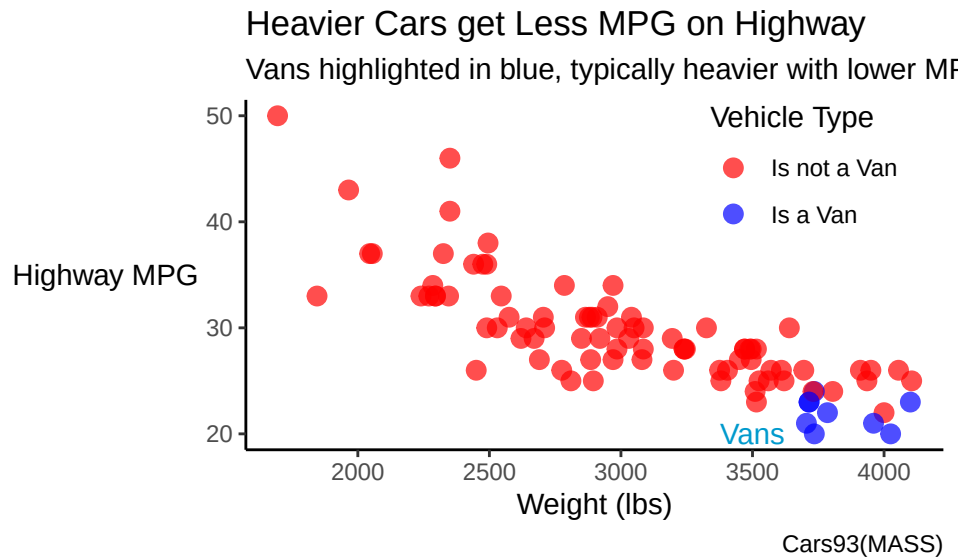
Color to Highlight

```
gf_point(MPG.highway ~ Weight , data = Cars93, color = ~(Type=="Van")) ) %>%gf_refine(annotate(geom = "t
```



#Revised, Point Size, Subtitle, Labels, Bold Title

```
gf_point(MPG.highway ~ Weight , data = Cars93, color = ~(Type=="Van"), size = 3, alpha = 0.7 ) %>%gf_re
```



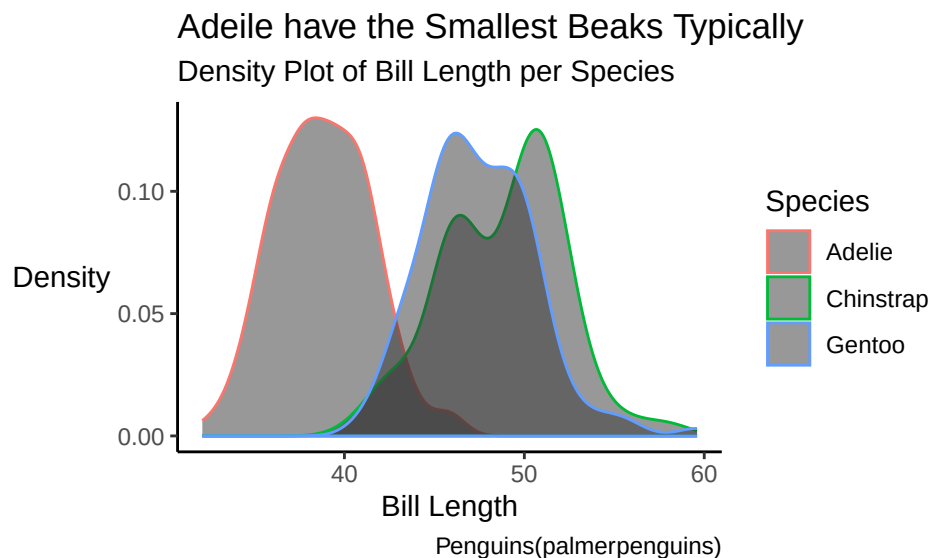
In the revised graph I updated the way the points look, I felt like a slightly bigger size would be beneficial for clarity but to compensate I decreased the alpha so the overlap becomes apparent. I also updated the subtitle to be more descriptive. The axes also had poor labels in the original. To rectify this issue, I gave them their proper units and labels.

Discrete Color

```
gf_density( ~ bill_length_mm , data = penguins , color = ~ species)%>%gf_refine( scale_fill_manual( val
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

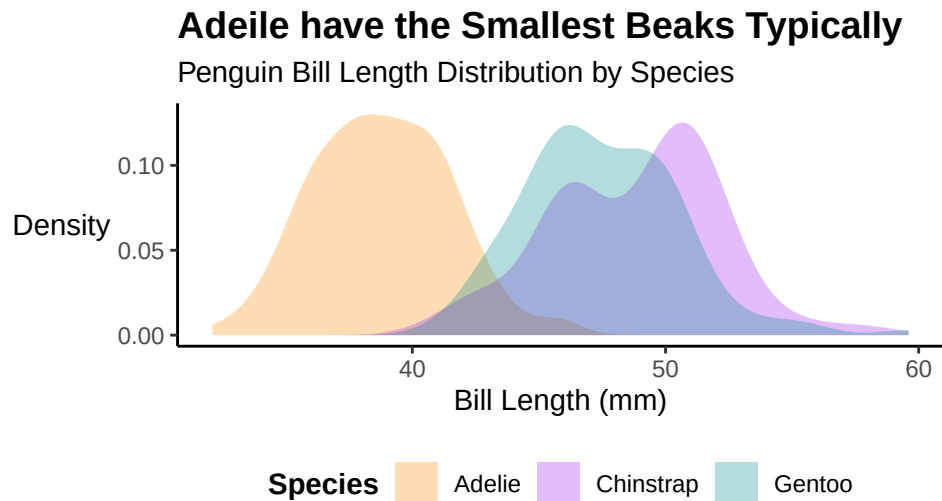
```
## (`stat_density()`).
```



```
#Revised Alpha, fill, Legend Bottom, plot and legend bold, Subtitle
```

```
gf_density( ~ bill_length_mm , data = penguins , fill = ~ species, alpha = 0.3)%>%gf_refine( scale_fill,
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).
```



Penguins(palmerpenguins)

One of the major points you mentioned with this one was to fill instead of color, taking that into account, I decided I would also use alpha levels to make it clear where there was overlap and how much their was. I also moved the legend to the bottom to make it look cleaner generally. The legend and title got bolding to draw attention to the important things to start on a graph. The final thing I changed was I added a more appropriate subtitle labeling it as distribution and not density for easier layman access.

Continuous Color

```
require(tidyr)
```

```
## Loading required package: tidyr
```

```
##
```

```
## Attaching package: 'tidyr'
```

```
## The following objects are masked from 'package:Matrix':
```

```
##
```

```
## expand, pack, unpack
```

```
df.volcano = as.data.frame(volcano) %>%
```

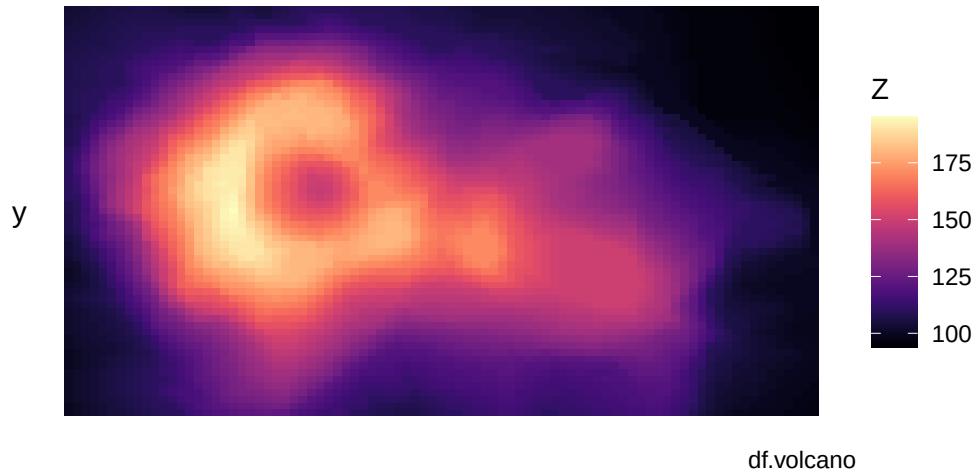
```
mutate(X = seq(from = 1, to = 87, by=1)) %>%
```

```
pivot_longer(cols = starts_with("V"), names_to = "Y", names_prefix = "V", values_to = "Z") %>%
```

```
mutate(Y = as.numeric(Y))
```

```
gf_raster(Z~X+Y, data = df.volcano)%>%gf_theme(theme_void())%>%gf_refine( scale_fill_viridis( option =
```

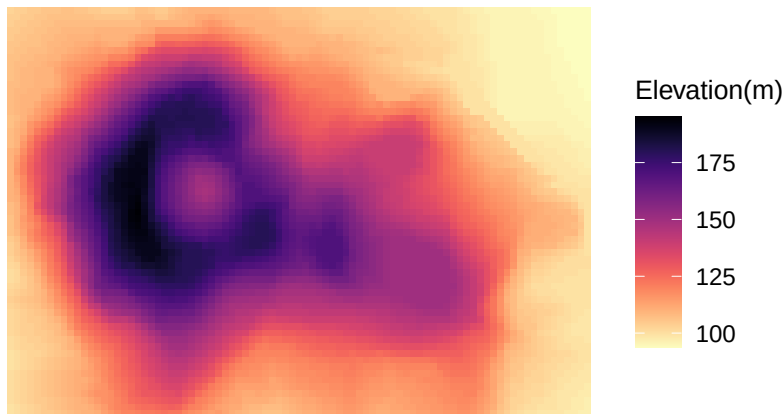
Volcano Plot



```
#Revised, Title, Reversed Direction, removed x and y, relabelled legend, fixed ratio
gf_raster(Z~X+Y, data = df.volcano) %>% gf_refine( scale_fill_viridis(direction = -1, option = "magma"
```

Volcano Topography Plot

Raster Plot of Volcano Elevation



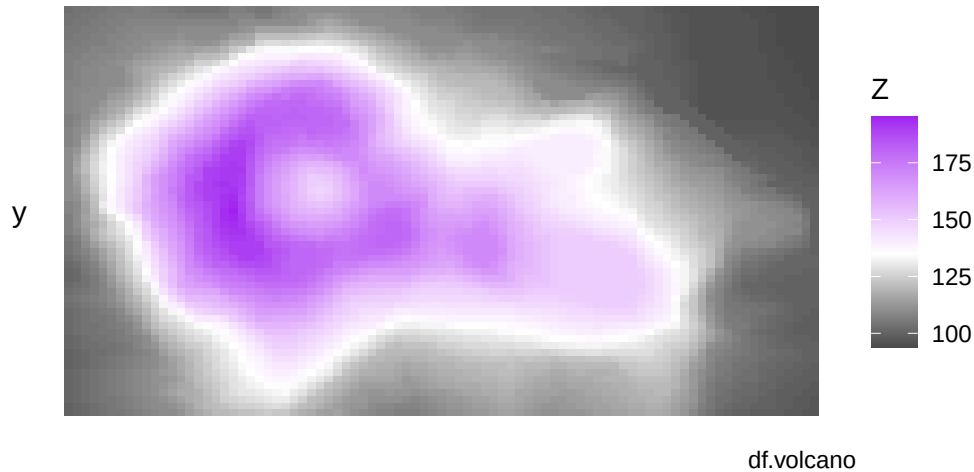
df.volcano

The Volcano Plot was kind of bland in the original plot. I felt like I needed to touch it up more in this. For starters, I prefer associating higher elevations with darker colors while the lower values are ‘empty’ so to speak and therefore closer to white. I liked the color scheme in magma though so I just reversed the direction. I also added topography to the title as I believed it would help better describe what we are looking at. In the subtitle I decided to include plot type, being raster. Additionally, I felt like x and y complicated things such that removing them was optimal. Additionally, I relabelled the key to be Elevation in terms of meters. then applied a fixed ratio.

Chosen “mid” point

```
gf_raster(Z~X+Y, data = df.volcano)%>%gf_theme(theme_void())%>%gf_refine(scale_fill_gradient2(low = "bl
```

Volcano Plot

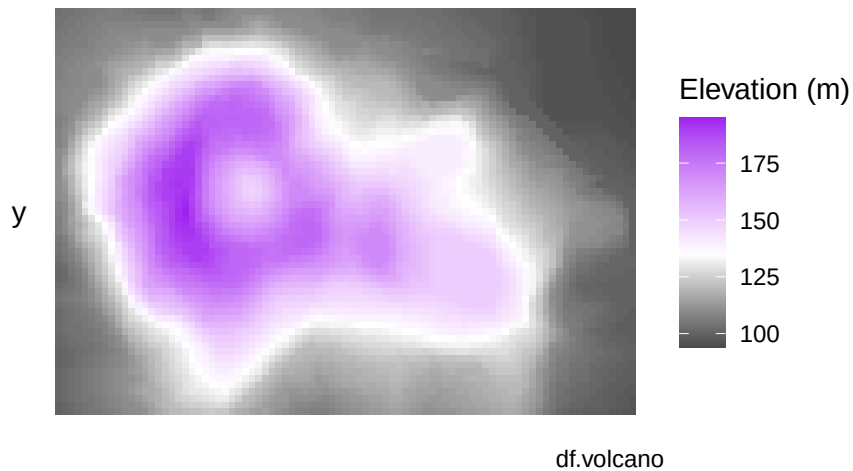


#Revised, Ratio, Subtitle, Legend

```
gf_raster(Z~X+Y, data = df.volcano)%>%gf_theme(theme_void())%>%gf_refine(scale_fill_gradient2(low = "bl
```

Volcano Plot

Black: Below 135m | White: 135m | Purple: Above 135m



For the chosen “mid” point I was unsure what to change precisely but I felt updating the legend was a good starting point. Additionally, making the subtitle more descriptive was important too. Finally, I fixed the ratio to complete this revision cycle.